

Venance Komu

+1 (607) 454-2132 | vck1@alfred.edu | in/venance-komu | github.com/venancekomu03-hub

EDUCATION

Alfred University

Alfred, NY

Bachelor of Science in Electrical Engineering, Mathematics - Double Major; Computer Science Minor

Expected May 2029

- **Relevant Coursework:** Circuit Theory, Digital Logic, Computer Organization & Architecture, Data Structures and Algorithms, Object-Oriented Programming (OOP), Linear Algebra, Differential Equations, Discrete Mathematics, Probability & Statistics
- **SAT Math:** 790/800 | **Honors:** Dean's List (2025, 2026)

TECHNICAL SKILLS

Programming Languages: Python, C, C++, MATLAB, VHDL, Java, SQL

Frameworks & Libraries: PyTorch, YOLOv8 (Ultralytics), OpenCV, TensorFlow/Keras, scikit-learn, NumPy, pandas, Matplotlib

Tools & Operating Systems: SolidWorks, LTspice, Linux (Ubuntu), macOS, Windows, Git, GitHub, Visual Studio, Excel

EXPERIENCE

Alfred University, Inamori School of Engineering

May 2026 – Jul 2026

Undergraduate Researcher

Alfred, NY

- Built a **Material Flow Analysis (MFA)** model of gallium, a critical **semiconductor** material, across secondary sources, using **Monte Carlo simulation** to quantify uncertainty in recovery for the **U.S. supply chain**.
- Analyzed **X-ray fluorescence (XRF)** measurements with **Python** scripts and **Matplotlib** plots to assess gallium concentrations and compare materials as potential recovery sources, versioning analysis code in **Git/GitHub**.
- Compiled and cleaned data on secondary gallium sources in **Python (pandas, NumPy)** and **Excel**, applying **descriptive statistics** to compare materials and assess their potential for recovery.

National Housing Corporation

Sep 2024 – Jan 2025

Intern

Moshi, Tanzania

- Analyzed property tax, payment, and maintenance records in **Excel** to identify discrepancies, modeling **trends over time** and **forecasting** from past records, and presenting findings in **PowerPoint** to support management decisions.
- Restructured and standardized property and tenant datasets in **Excel** and **SQL**, improving data accuracy, record retrieval, and operational efficiency across property-management processes.

TECHNICAL & PROFESSIONAL DEVELOPMENT

CodePath

Spring 2026

Technical Interview Prep Student

Remote (San Francisco, CA)

- Completed intensive training in **data structures and algorithms** and computational problem-solving in **Python**.
- Implemented **Python** algorithms using **recursion, sorting, searching, hash tables**, and linked data structures.

freeCodeCamp

Fall 2026

Machine Learning & Scientific Computing with Python

Remote (Alfred, NY)

- Built **TensorFlow/Keras** models for image classification (**CNN, computer vision**), SMS spam detection (**NLP**), and healthcare cost prediction (**regression**).
- Developed a **K-Nearest Neighbors (KNN)** book recommender with **scikit-learn** on a **1.1M-rating** dataset, a Rock Paper Scissors game-playing algorithm, and **object-oriented** Python programs.

PROJECTS

Critical Materials Circuit Component Detector | *Python, PyTorch, YOLOv8, OpenCV, NumPy/pandas, Linux* Summer 2026

- Trained a **YOLOv8 computer vision** model in **PyTorch** to detect 5 component classes (ICs, capacitors, resistors, transistors, connectors) on about **500 PCB images** with **2,000+ labeled instances**, split 70/15/15.
- Matched detections to a **critical-materials** reference database to estimate each board's material content (**mg or g per PCB**), stating the assumptions behind each estimate.

Collaborative Food Cargo Tracking System – WFP Tanzania | *Human-Centered Design, Figma*

Aug 2024 – Jan 2025

- Designed a **real-time dashboard** integrating **geofencing, GPS tracking**, and **AI-powered metrics analysis**, projected to reduce communication inefficiencies and emergency calls by **80%**, enabling faster delivery of food aid to vulnerable communities.

Tiny House Project | *Engineering Project, Python, Excel*

Sep 2025 – Present

- Evaluated and optimized **battery energy-storage** configurations for an off-grid tiny house, analyzing capacity, efficiency, load requirements, and **cost estimates** in **Python** and **Excel** to support reliable **power delivery**.

LEADERSHIP & ACTIVITIES

Robotics Club, Alfred University | *Treasurer*

Sep 2025 – Present

- Allocate the club's budget across **robotics projects** and activities, maintaining its financial records in **Excel**.

ColorStack, Alfred University Chapter | *Treasurer*

Aug 2026 – Present

- Build semester budgets for workshops, mentorship, and career events; track expenses in **Excel** and prepare funding requests.